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Insuring Unemployment Out of Your Own Pension*

Causal Evidence from Mexico's Retiro Parcial por Desempleo

Esteban Degetau

Instituto Mexicano del Seguro Social, Barcelona School of Economics

estebandegetau@gmail.com

Abstract I study the labor market effects of Mexico's Retiro Parcial por Desempleo (RPD), which allows unemployed workers to withdraw pension savings. Using rich administrative data and a fuzzy regression discontinuity design exploiting a sharp eligibility threshold, I first show that eligibility increases program take-up by 3.6 percentage points twelve months after displacement. Leveraging these take-up data, I estimate the local average treatment effect on compliers and find that RPD use prolongs time out of formal employment by 36 weeks over three years, without gains in reemployment wages, job stability, or cumulative formal earnings. Testing differences between subgroups directly, the cost concentrates on workers with below-median prior earnings and on men; the analogous gradients by age and by pandemic exposure are suggestive but not statistically distinguishable. No subgroup shows a detectable gain in three-year cumulative earnings, and the only reemployment-wage gain significant at the 5% level accrues to workers already above the median. Because the additional weeks are weeks without pension contributions, the cost compounds until retirement: the loss in accumulated balance is several times the sum withdrawn, and roughly two fifths of it stems from the induced unemployment rather than from the withdrawal itself. The results suggest that while RPD provides short-term liquidity, it does not improve long-term outcomes and may exacerbate existing inequalities.

Keywords: unemployment insurance, income substitution, Mexico, labor economics

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1 Introduction

Job loss can cause severe income disruptions, yet in the absence of public intervention, workers must self-insure by borrowing, drawing down savings, or relying on informal support networks. Economists have long argued that such self-insurance is inefficient: when credit markets are incomplete and workers are risk-averse, precautionary strategies often fall short, and the welfare losses from income volatility can be large. Unemployment insurance (UI) programs address this market failure by pooling risk across individuals and time, thereby enabling workers to smooth consumption during periods of joblessness (Baily 1978; Gruber 1997). Beyond their insurance function, UI programs may also improve job match quality by reducing the urgency to accept the first available job, leading to higher earnings and long-run productivity (Acemoglu and Shimer 1999; 2000).

These benefits, however, must be balanced against potential moral hazard; generous benefits may reduce the urgency to search for work, prolonging unemployment. Chetty (2008) showed that this behavioral response in fact decomposes into two mechanisms with very different welfare content: a moral hazard component, in which the benefit distorts the incentive to search, and a liquidity component, in which it relaxes a constraint that had been forcing the worker to accept the first available offer. Only the first is an efficiency cost, and the two are difficult to separate because in most programs they move together.

In developing countries, where fiscal capacity is limited and informality widespread, traditional UI systems are often unviable. As a result, many governments rely on self-financed alternatives designed to provide limited liquidity while minimizing work disincentives. One such program is Mexico's Retiro Parcial por Desempleo (RPD), which allows unemployed formal-sector workers to withdraw a portion of their pension savings upon job loss. Because the worker bears the entire cost of the benefit through a smaller pension, with no risk pooling and no fiscal outlay, conventional reasoning predicts a muted behavioral response — the property individual unemployment accounts were designed to deliver (Feldstein and Altman 2007), and the one Chile's system appears to exhibit, where the response concentrates in the solidarity-funded tranche rather than in the worker's own account (Hartley et al. 2011). The RPD is therefore an unusually clean setting in which to ask whether that prediction holds.

This paper addresses the question: What are the causal labor market effects of access to the RPD program in Mexico? I study the impact of RPD eligibility and take-up on three key outcomes: (i) job search behavior, proxied by unemployment duration and survival in unemployment; (ii) job quality, measured by reemployment wages and job duration; and (iii) medium-term labor market performance, defined as formal earnings and months worked in the three years following displacement. Understanding these effects is critical for assessing whether quasi-UI schemes like RPD enable better job matches or merely delay reemployment, and whether they reinforce or mitigate longer-term income risks.

A growing body of research has examined the effects of UI on labor market outcomes in both high- and low-income contexts. In advanced economies, natural experiments and regression discontinuity designs show that more generous UI consistently prolongs unemployment spells (Katz and Meyer 1990; Schmieder et al. 2012). Evidence on job quality is more mixed: while some studies report gains in reemployment wages and job stability (Nekoei and Weber 2017; Centeno and Novo 2006), others find negligible or even adverse effects on earnings (Schmieder and Wachter 2016; Le Barbanchon 2016). In middle-income countries, UI interacts with informality in complex ways. For instance, Britto (2022) finds that in Brazil, over half of the additional unemployment induced by extended UI is offset by increased informal employment, raising concerns about behavioral leakage and program effectiveness.

This study contributes to the literature by being the first to estimate the causal impact of Mexico's RPD program using administrative data and a fuzzy regression discontinuity design. While earlier causal studies (Velázquez Guadarrama 2023; Carreño Godínez 2025) relied on survey data, I leverage high-frequency longitudinal records from the Mexican social security system to construct a clean analytic sample and directly observe eligibility, take-up, and formal labor market trajectories. The RPD's eligibility threshold—requiring at least two years of social security contributions—offers a compelling quasi-experimental setting. By exploiting this discontinuity and tracking administrative take-up, I estimate local average treatment effects (LATEs) for compliers across a rich set of labor market outcomes. While internal validity is strong, generalizability is limited to younger formal-sector workers near the eligibility threshold.

The results show that RPD take-up has large and significant effects on unemployment duration: users remain out of formal employment for approximately 36.3 additional weeks over a three-year period, from a baseline of 51.5 weeks for the ineligible, with effects evident within the first three months after displacement. However, these extended spells do not translate into higher job quality. RPD users experience no detectable gains in wages, job duration, or cumulative earnings, and in no subgroup is the effect on three-year cumulative earnings distinguishable from zero. The only reemployment-wage gain that reaches significance at the 5% level accrues to workers whose prior earnings were already above the median. These findings challenge optimistic models of UI that predict gains in productivity and matching quality (Acemoglu and Shimer 2000), and instead suggest that self-financed programs like RPD may distort job search without delivering meaningful returns in terms of labor market outcomes.

Heterogeneity analyses locate that cost rather than dispersing it. Testing the difference between subgroups directly — rather than comparing the magnitude of separately estimated coefficients — the duration cost falls on workers with below-median prior earnings and on men, and in both cases the gap between the two halves of the split is itself statistically significant. Splits by age and by exposure to the COVID-19 pandemic point in the same direction, but their differences cannot be distinguished from zero, and I report them as suggestive rather than as findings. The pattern that survives testing is therefore one in which the program's cost concentrates among workers who arrive at unemployment with the least to fall back on, while no group is compensated by better subsequent employment.

These findings offer several policy-relevant insights. First, the structure of the RPD program—being self-financed and actuarially neutral—appears to reduce concerns about fraud and fiscal burden but also limits its redistributive potential. Second, the regressive nature of its benefits—favoring those with more savings and stronger labor market histories—raises concerns about equity and adequacy. Lastly, while the program may be appropriate as a minimal safety net, it does not substitute for the functions of traditional unemployment insurance. The results contribute empirical grounding to policy debates in Mexico and other developing economies over whether to expand self-financed schemes, replace them with publicly funded UI, or design hybrid systems tailored to institutional capacity.

Nonetheless, the study has limitations. The fuzzy RD design identifies local treatment effects for a narrow group of workers—those displaced in their third year of formal labor market participation—limiting external validity. Moreover, the administrative data observe formal employment and only formal employment, so a worker who moves into informal or self-employment is indistinguishable from one who continues searching; Section 5.5 confronts what that means for the interpretation of the duration result. Finally, the modest first stage—a 3.6 percentage point increase in take-up—means the estimates describe a small group of compliers and constrains precision in the second stage, necessitating cautious interpretation.

The remainder of the paper proceeds as follows. Section 2 provides institutional background on the RPD program and Mexico's broader unemployment protection framework. Section 3 describes the data and sample construction. Section 4 outlines the approach. Section 5 presents the main findings, including heterogeneity analysis. I finish with a discussion of the findings and their implications for policy.

2 Institutional Background

This section provides an overview of the institutional framework surrounding unemployment protection in Mexico, with a focus on the *Retiro Parcial por Desempleo* (RPD) program. I begin by describing the eligibility conditions and usage rules of the RPD, a mechanism that allows unemployed workers to make partial withdrawals from their individual pension accounts. I then document the evolution of the program's adoption over time, highlighting its increased relevance during periods of economic crisis. Finally, I situate RPD within the broader context of income substitution in Mexico, where the failure to enforce legally mandated severance pay has rendered employer-based protections largely ineffective. In this setting, RPD emerges as a self-financed, readily accessible—but ultimately limited—alternative for workers facing job loss.

2.1 The Partial Withdrawal

The RPD program works as follows. The pension system in Mexico is a standard defined contributions scheme with individual accounts. While working in formality, workers accumulate resources financed in part by themselves, by their employer and by the government.

2.1.a Eligibility Criteria

If a worker faces an unemployment spell of at least 46 days, she may withdraw a fraction of her individual account balance through the RPD. The amount of money a worker can have access to depends on the number of years since her account was opened and the number of weeks she has contributed to social security. The *Ley del Seguro Social* considers two eligibility schemes, each allowing a worker to withdraw an amount up to:

A. Ten months of the minimum wage or one month of her most recent wage, whichever is lower, for workers satisfying:

i. having entered the formal labor market at least three years ago, and

ii. having contributed to social security for at least two years.[2]

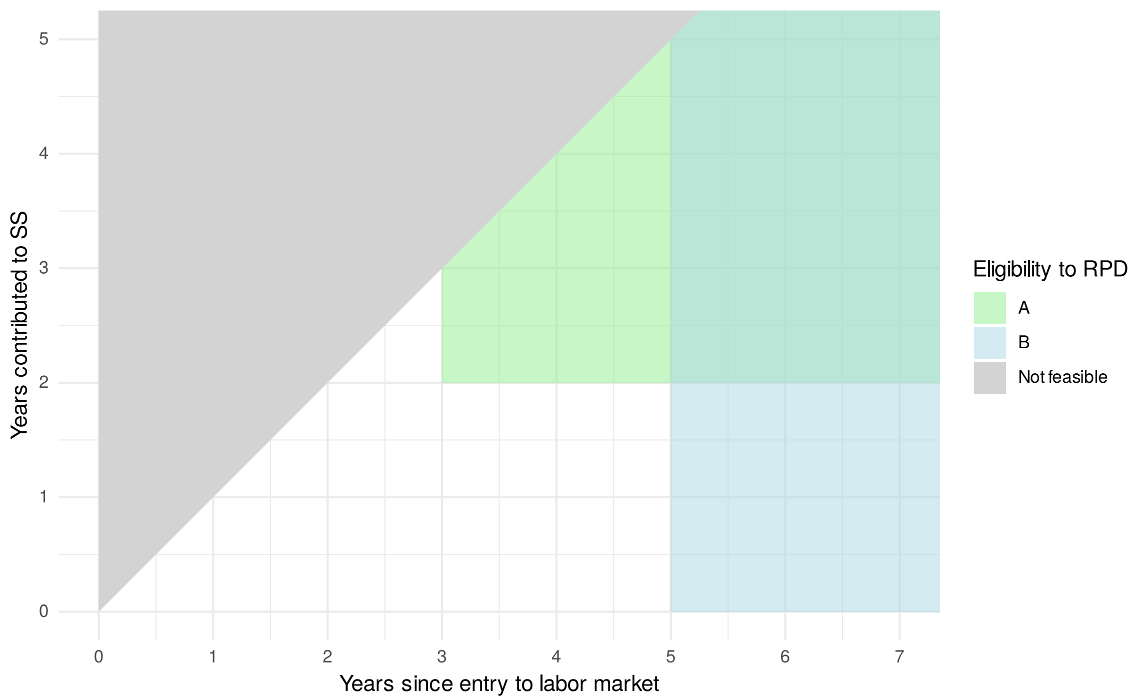
B. Three times her monthly wage or 11.5% of her account balance, whichever is lower, for workers who entered the formal labor market at least five years ago, regardless of contributions to social security.

Figure 1 provides a visual representation of the eligibility criteria. Note that a worker who became dismissed after four years since she entered the labor market, and with only one year of contributions to social security would not be immediately eligible to withdraw funds from her account. However, if she just waited in unemployment for a year, she would become eligible through Scheme B, without having to make further contributions. Hence, ineligible workers may become eligible by the mere passage of time. This feature of the eligibility structure is important to the construction of the analysis sample, as I show in Section 3.2.

Moreover, workers who are eligible through both schemes may choose the one that allows them to withdraw the largest amount. Having used the RPD once, a worker may use it again, only after waiting for five years since her last withdrawal. Importantly, using the RPD not only depletes the worker's individual pension savings account, but it also reduces proportionally the amount of days contributed to social security, which are used to determine and finance the pension at retirement age. This is a key feature of the RPD, as it allows workers to access their savings during unemployment, but it also reduces their future pension benefits.

[2]Contributions to social security need not be continuous to reach eligibility to RPD.

Figure 1: Eligibility Criteria



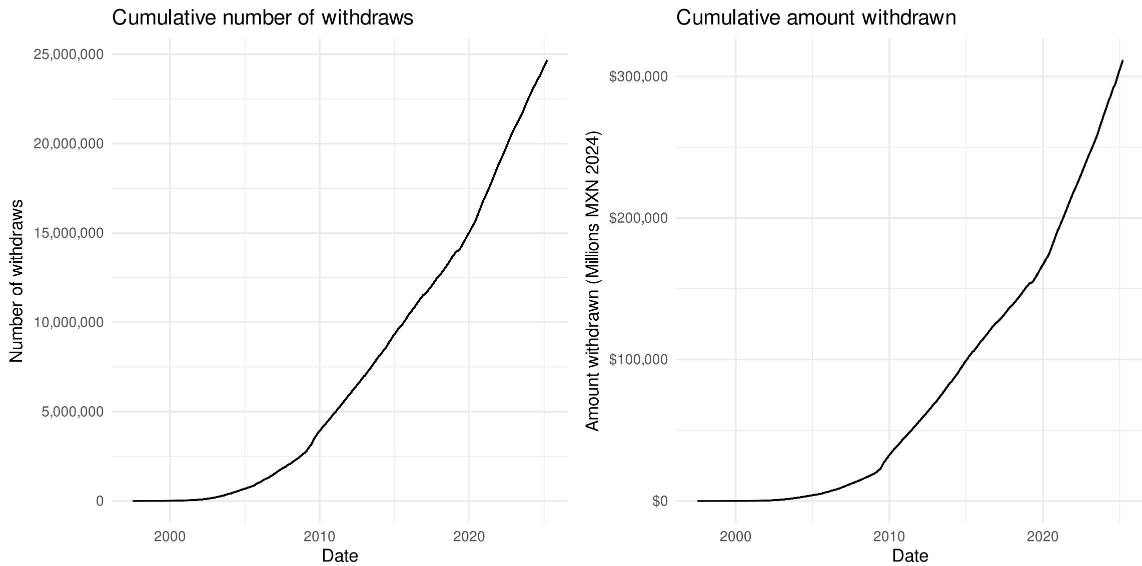
Note: This figure conceptualizes eligibility to RPD. The x-axis shows the number of years since entry to the labor market, while the y-axis shows the number of years contributed to social security. The green area represents the eligibility criteria for Scheme A: having entered the formal labor market at least three years before and having contributed to social security for at lest two years. The blue area represents the eligibility criteria for Scheme B: having entered the formal labor market at least five years before, regardless of contributions to social security. The gray area represents the not feasible zone: workers cannot contribute more days to social security than they have been in the formal labor market.

2.2 Usage and Adoption

Figure 2 shows the cumulative number of partial withdraws from the individualized pension accounts since it was implemented in 1997. The data comes from a confidential administrative data set from the *Instituto Mexicano del Seguro Social* (IMSS). The RPD program was first implement in 2009 to give unemployed workers a source of income substitution in the context of the Great Recession. This event can be seen as a sharp increase in the usage of RPD, but also as a change in the trend of the number of withdrawals from the pension system. Accordingly, in the next big employment crisis, the COVID-19 pandemic, another trend increase followed. This shifts in the slope of the cumulative usage of RPD shows that employment crises have a persistent effect on the usage of the program, likely due to the increased salience of the program.

To put figures into perspective, by the end of 2024, 16,966,851 workers had made 24,598,838 withdrawals from the pension system. By that same moment, IMSS reported a total of 22,238,379 active workers in the formal labor market. Indeed, not all RPD users were active workers by the end of 2024, but this number gives a sense of the magnitude of the usage program.

Figure 2: Usage and Adoption



Note: This figure shows the cumulative number of partial withdrawals (left) and the cumulative amount withdrawn (right) out of the Mexican individual pension account system (Afore). Source: IMSS.

2.3 Income Substitution for the Unemployed in Mexico

In theory, Mexican labor law provides mandatory severance pay (indemnización constitucional) as the main form of income substitution for workers who lose their jobs. Upon unjustified dismissal, employers are required to provide affected workers with three months of salary, plus seniority bonuses and accrued benefits. This legal severance framework reflects the broader Latin American tradition of employer liability systems as a substitute for unemployment insurance. However, in practice, enforcement of these entitlements is weak, uneven, and often illusory—rendering severance rights largely ineffective as a tool for income smoothing during unemployment.

The core issue lies in the functioning of Mexico's labor courts. In particular, in Mexico City, workers who wish to claim their legally mandated severance must navigate a judicial process that is slow, opaque, and vulnerable to both corruption and administrative incompetence. Degetau (2023) provides experimental evidence from the Mexico City Local Labor Courts showing that the mere act of serving notification—a basic procedural step that triggers the legal process—is often compromised. The study reveals that notification orders may fail not due to complexity or distance but because of strategic delays enabled by collusion between defendants (employers) and court notifiers.

Further compounding these problems is the widespread information asymmetry among litigants. Sadka et al. (2024) demonstrate that plaintiffs in labor disputes often have unrealistic expectations about the outcomes of their cases, which leads to protracted litigation and missed opportunities for settlement. Their randomized controlled trial shows that providing plaintiffs with predictive information about expected case duration, probability of success, and likely monetary outcomes significantly increases same-day settlement rates and improves short-term financial well-being. However, this intervention also reveals the deep dysfunction of the system: only when plaintiffs are given external tools to navigate uncertainty can they extract meaningful value from the legal process.

The statistical odds of a dismissed worker successfully enforcing their severance rights are vanishingly low. As Sadka et al. (2024) document, more than half of all dismissed workers in Mexico report never receiving severance pay, yet only 15% pursue legal action—a striking indication of the widespread perception that litigation is futile . And with good reason: nearly 30% of lawsuits remain unresolved four years after being filed, leaving workers in legal limbo with no income and no resolution. Even among the rare plaintiffs who persist through the full litigation process and win a favorable judgment, only 20% ever collect the full amount awarded. The practical message is devastatingly clear: in Mexico's current labor justice system, even winning your case is no guarantee of receiving what the law promises. For the average worker facing dismissal, the path to severance is not just slow and bureaucratic—it is overwhelmingly stacked against them.

Taken together, these findings paint a troubling picture: while Mexico's legal framework nominally guarantees severance pay as income support for the unemployed, the institutions tasked with enforcing this right often fail to deliver. Procedural inefficiencies, the potential for corruption, and severe information gaps all contribute to a de facto collapse of enforcement. As a result, many dismissed workers are left without effective legal recourse, and severance pay—despite being codified in law—cannot be relied upon as a real source of income during unemployment.

In this vacuum, workers may increasingly turn to alternative, self-financed mechanisms such as the RPD program—a partial withdrawal from their individual pension accounts—as their only accessible form of income substitution. Unlike severance pay, RPD is automatic and does not require litigation. However, it also lacks redistributive features, may deplete retirement savings, and provides only modest relief. This shift from legal entitlements to self-funded liquidity illustrates a broader institutional failure: in the absence of credible enforcement, even well-designed labor protections may become functionally irrelevant.

3 Data

3.1 Data Sources

To study the effect of the Retiro Parcial por Desempleo (RPD) on labor market outcomes, I use two confidential administrative data sources. The first is a matched employer-employee panel from the Instituto Mexicano del Seguro Social (IMSS), which tracks the entirety of the formal labor market. From this data set, I construct eligibility criteria and labor market outcomes. I restrict the analysis

[3]This restriction ensures that all workers are subject to the same pension reform (in effect since July 1997) and thus face uniform incentives regarding the accumulation of weeks of contribution and the use of their individual pension account balance for income substitution during unemployment.

to workers who first entered the formal labor market between January 2010 and December 2017.[3] This yields a universe of 14,819,690 workers across 406,813,240 labor contracts.

Although the IMSS employment data does not directly record unemployment episodes (i.e., the cause, onset, or duration of unemployment), unemployment spells can be inferred from gaps between labor contracts. I define a gap as a period during which an individual is not formally employed by any firm. While such gaps do not necessarily imply unemployment[4], they are precisely the criterion the RPD authority uses to determine whether a worker is eligible for benefits.

It is worth fixing the exact scope of this variable at the outset, because it governs the interpretation of every result that follows. What these data observe is time spent **without registered formal employment**, not unemployment in the sense of a labor force survey. For economy of language I refer to it as "unemployment duration" throughout, but a worker who takes a job in the informal sector, or who begins working on her own account, appears in these records exactly as one who remains without work and searching. The two are indistinguishable here. Section 5.5) examines what that distinction implies for the interpretation of the main results, and shows which of the paper's conclusions depend on resolving it and which do not.

The second data source is the administrative registry of partial withdrawals from individual pension accounts. This registry includes every instance of RPD usage and records the date of withdrawal, amount withdrawn, account balance, weeks contributed to social security at that point, the last employer's ID, and the worker's unique identifier (Número de Seguridad Social, NSS). Crucially, I use the NSS to match the two data sets, enabling me to track the labor market trajectories of RPD users before and after its use. See Section 2.2 for a detailed overview of the RPD program and its adoption.

3.2 Sample Selection

To estimate the effect of RPD on labor market outcomes, I leverage the eligibility criterion of having at least two years of contributions to social security at the moment of unemployment as s source of quasi-exogenous variation, (See Section 2 for context on the eligibility criteria.). Hence, I restrict the analysis sample to workers who (i) faced exactly one unemployment spell of at least 46 days during the third year since they joined the formal labor marked, and who (ii) had accumulated between one and three years worth of contributions to the social security system.

Additionally, I restricted the analysis sample to workers with (a) ages between 18 and 65 years old, (b) who were registered in the *Régimen obligatorio* as permanent or eventual workers, (c) were not self-employed, (d) were not registered to an agrarian firm, and (e) were not in the top 1% of the unemployment duration distribution.

These restrictions yield an analysis sample of 877,749 workers. Figure 3 shows the sample selection of workers in the eligibility criteria space at the time their unique unemployment spell began. Although the restrictions I impose on the analysis sample render my results non-generalizable to the entire population of workers, they allow me to leverage the institutional characteristics

[4] A gap could result from voluntary separation, a transition into informal work, or involuntary displacement.

for identification of the effect of eligibility to RPD on labor market outcomes. Table A.1 shows additional summary statistics of the analysis sample.

Figure 3: Sample Selection

Note: This figure provides a conceptual representation of eligibility for the RPD program. The x-axis denotes years since entry into the formal labor market, and the y-axis indicates years of contributions to social security. The green area corresponds to Scheme A eligibility, which requires at least three years since labor market entry and a minimum of two years of social security contributions. The blue area corresponds to Scheme B eligibility, which requires at least five years since labor market entry, irrespective of contribution history. The red square marks the subsample selected for empirical analysis. The gray area represents an infeasible region, as individuals cannot accumulate more years of social security contributions than years since formal labor market entry.

Within the analysis sample, only those workers who had accumulated at least two years of contributions to the social security system by the time they become displaced are eligible to withdraw funds from their pension account. Table 1 shows workers' features across eligibility status at the moment of displacement. Indeed, there is a systematic positive selection of workers into eligibility, as workers who are eligible have contributed more to social security, had higher wages and longer tenures before displacement. This is expected and is not a threat to identification: the comparison that matters is not between all eligible and all ineligible workers, but between workers just on either side of the threshold. Critically for my identification strategy, I show in Section 4.1 that these differences are smooth around the eligibility threshold of two years of contributions, and that no predetermined covariate exhibits a discontinuity there once the multiplicity of tests is accounted for.

Table 1: Balance of Outcomes and Covariates by Eligibility Status

Characteristic	**Eligible** N = 427,927	**Ineligible** N = 449,822	**p-value**
Displacement date	2,018.07 (2.12)	2,018.02 (2.10)	<0.001
Outcomes			
Take up - 12 months	37,400 (8.7%)	685 (0.2%)	<0.001
Survival - 3 months	322,651 (75%)	366,208 (81%)	<0.001
Unemp. Dur. - Cens. 6 months	20.0 (7.2)	21.2 (6.7)	<0.001
Unemp. Dur. - Cens. 36 months	50 (48)	56 (50)	<0.001
Eligibility criteria			
Days since entry to formal labor market	1,264 (104)	1,265 (103)	<0.001
Days contributed to SS	914 (106)	545 (106)	<0.001
Previous job			
Tenure (weeks)	35 (35)	21 (21)	<0.001
Daily wage (MXN 2024)	283 (199)	248 (151)	<0.001
Demographics			
Female	163,078 (38%)	167,778 (37%)	<0.001
Age at displacement date (years)	24.9 (5.6)	24.5 (5.3)	<0.001
No CURP	13,929 (3.3%)	16,106 (3.6%)	<0.001

Note: This table reports means and standard deviations (in parentheses) for key characteristics in the analysis sample, disaggregated by eligibility status at the date of displacement. For binary variables—such as *Take-up*, *Survival*, Female, and *No CURP*—the table reports the count and the corresponding percentage (in parentheses). *No CURP* indicates that the worker does not have a unique population identification number (CURP) registered with the IMSS. The p-value column reports the results from hypothesis tests of individual differences of means across eligible and ineligible groups.

4 Empirical Strategy

To estimate the causal effect of the *Retiro Parcial por Desempleo* (RPD) on labor market outcomes I use a regression discontinuity design (RDD) exploiting the eligibility condition of having contributed to social security for at least two years at the moment of displacement. I leverage data on RPD take up to estimate the local average treatment effect (LATE) on compliers using a fuzzy discontinuity design. The fuzzy RDD can be expressed as a two-stage least squares regression, where the first stage estimates the effect of eligibility on take up of the RPD program as in

$$T_i = \alpha + \beta D_i + f(C_i - 2) + u_i \quad \text{where} \quad \mid C_i - 2 \mid \leq h \tag{1}$$

where T_i is a dummy indicating worker i's take up to RPD, D_i indicates whether i is eligible to withdraw funds from her pension account, i.e. $C_i \geq 2$, where C_i represents i's contributions to SS measured in years. $f(\cdot)$ is a polynomial function of C_i and h is the bandwidth around the threshold. β identifies the causal effect of eligibility on take up of the RPD program.

Hence, the second stage estimates the effect of taking up RPD on labor market outcomes as in

$$Y_i = \gamma + \tau \hat{T}_i + f(C_i - 2) + v_i \quad \text{where} \quad \mid C_i - 2 \mid \leq h \tag{2}$$

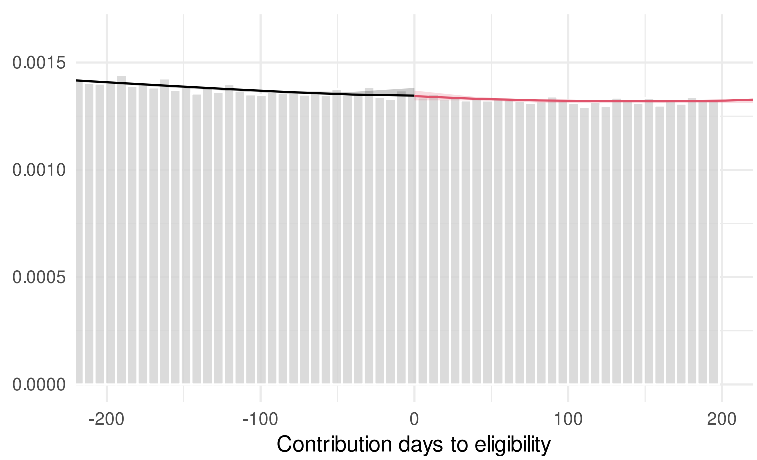
where Y_i is an outcome of interest, $\hat{T}_i$ is the predicted value of the first stage. The main parameter of interest is τ, which identifies the LATE around the eligibility threshold. I use a local linear polynomial ($p = 1$) with a triangular kernel and the bandwidth selected using the method proposed by Calonico et al. (2020). Specifically, I'm interested in the hypothesis test given by $H_0 : \tau = 0$ vs $H_1 : \tau \neq 0$. For inference, I use heteroskedasticity-robust standard errors with nearest-neighbor variance estimator, as proposed by Calonico et al. (2014).

4.1 Evidence on Design Validity

I provide two main tests on the validity of the sharp design in (1) plus an additional test on the validity of the fuzzy design in (2). First, I show that there is no evidence of manipulation around the eligibility threshold. Second, I show that the covariates are balanced around the eligibility threshold. Finally, I show that the first stage is strong.

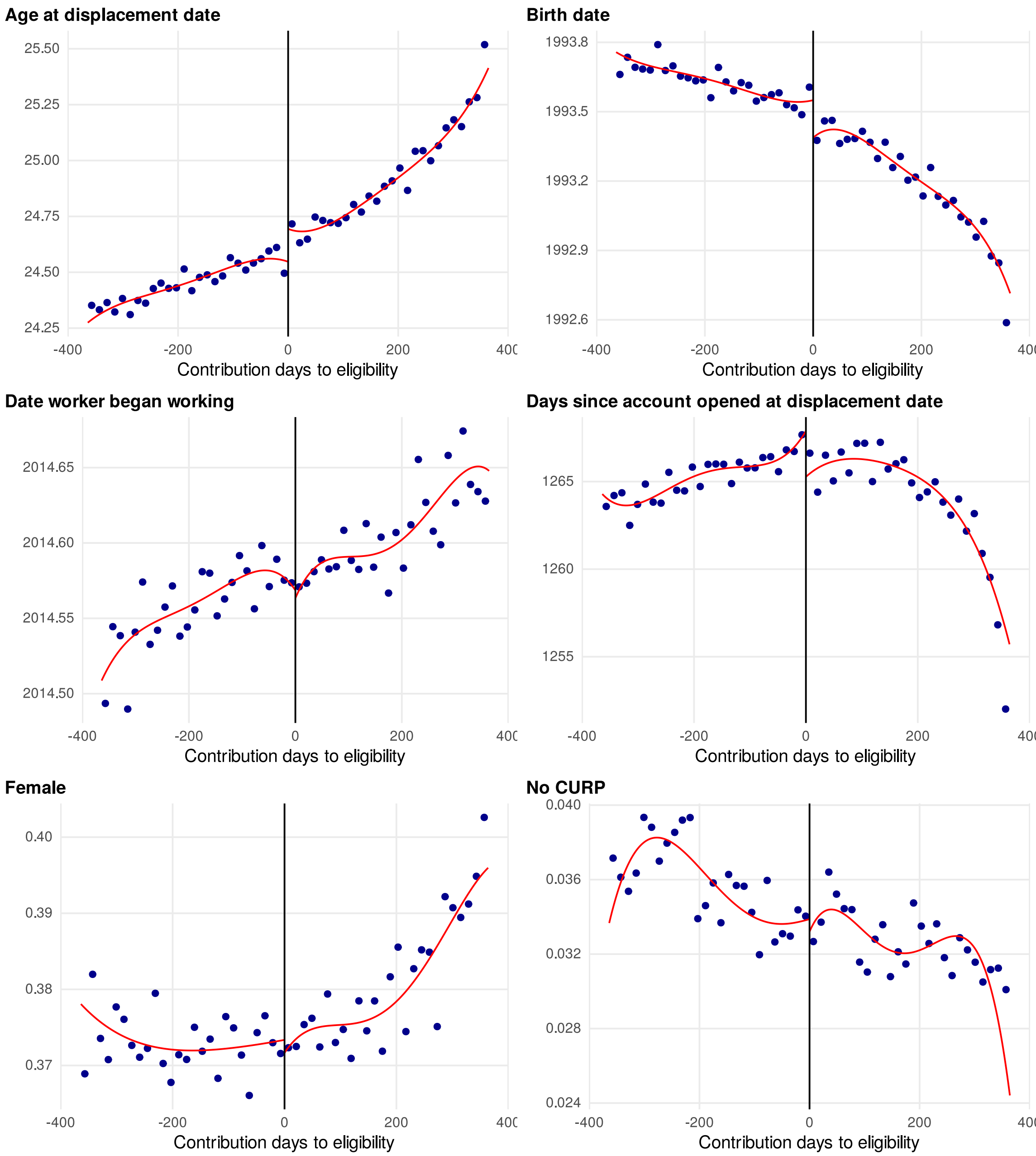
First, Figure 4 shows the result of the Cattaneo et al. (2020) test for manipulation of the running variable. The figure shows no significant bunching around the eligibility threshold, consistent with the absence of manipulation. Second, Figure 5, Figure 6 show the continuity of covariates of workers around the eligibility threshold. Crucially, Table 2 shows the estimation and inference of (1), showing the covariates do not jump across the eligibility threshold systematically (Cattaneo et al. 2019). Finally, I present evidence on the strength of the first stage in the Section 5.1.

Figure 4: Density of the Running Variable



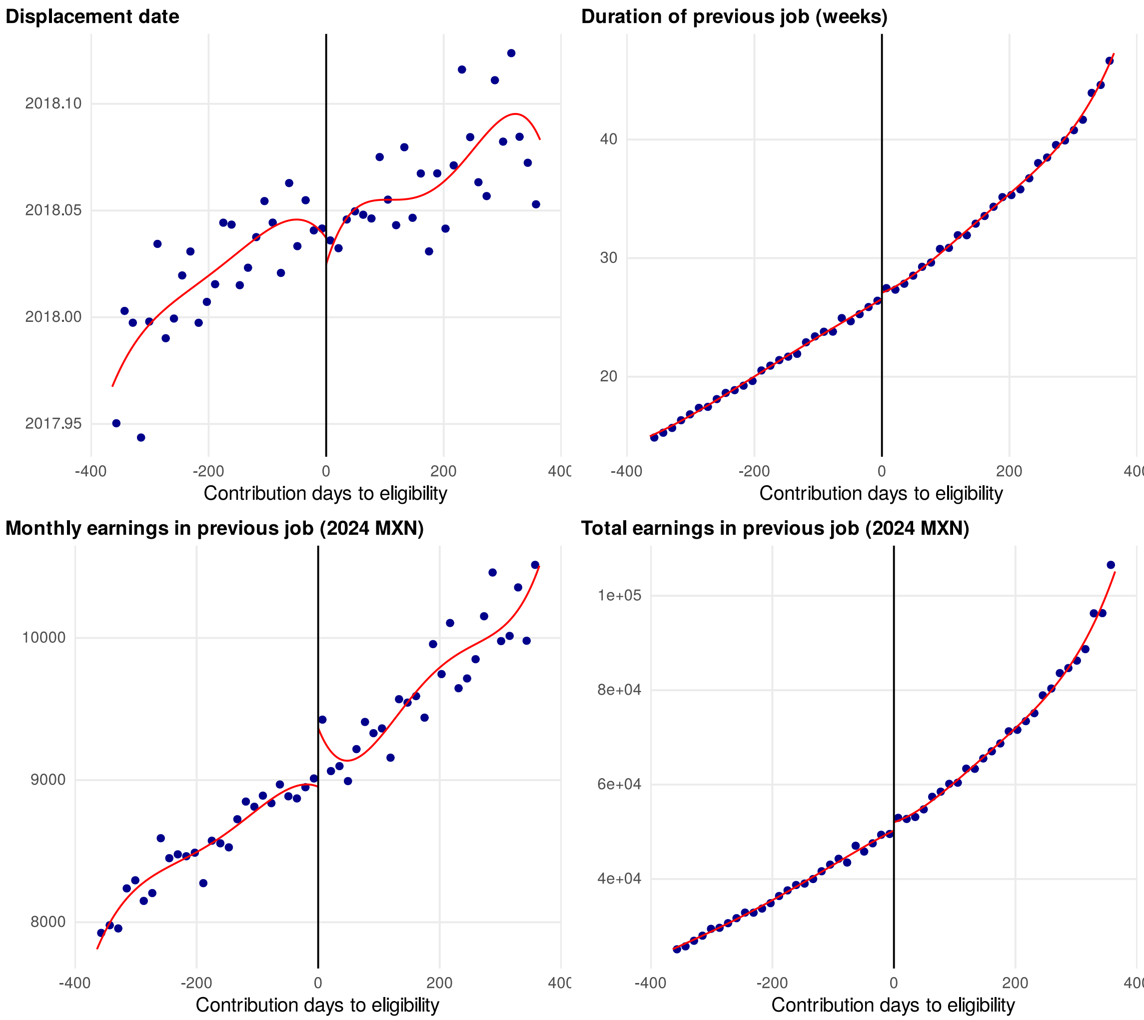
Note: This figure shows there was no manipulation into eligibility by altering the days of contributions to social security at the moment of displacement (Cattaneo et al. 2020).

Figure 5: Covariates around Eligibility Threshold



Note: This figure shows (*i*) sample means for evenly-spaced 14-day bins relative to the eligibility threshold, and (*ii*) the underlying regression function estimated using a triangular kernel, optimal bandwidth and a linear local polynomial using Calonico et al. (2015). The figure shows workers' covariates do not systematically jump at the eligibility threshold.

Figure 6: Previous Job Outcomes around Eligibility Threshold



Note: his figure shows (*i*) sample means for evenly-spaced 14-day bins relative to the eligibility threshold, and (*ii*) the underlying regression function estimated using a triangular kernel, optimal bandwidth and a linear local polynomial using Calonico et al. (2015). The figure shows workers' previous job outcomes do not systematically jump at the eligibility threshold.

Table 2 reports the formal continuity tests for the ten predetermined covariates. Three of them reject continuity at the 5% level individually, but these p-values are not the right basis for a conclusion: ten simultaneous tests are being conducted, and under the null one would expect roughly one rejection at 5% by chance alone. Applying the Holm step-down correction, which controls the probability of at least one false rejection without assuming independence across covariates, **no covariate retains significance at the 5% level** (Aickin and Gensler 1996). The correction matters here in particular because two of the three individually significant covariates — age and date of birth — are mechanically the same test with opposite sign, and so are perfectly dependent.

The discontinuities that are individually detectable are also economically negligible. The jump in age amounts to about a month and a half against a sample mean of roughly twenty-five years, and the jump in days since the pension account was opened is on the order of two days against a mean above twelve hundred. Neither is of a magnitude that could plausibly generate the effects reported below.

Table 2: Continuity of Predetermined Covariates at the Eligibility Threshold

Covariate	RD effect	Std. err.	p	p (Holm)	Eff. N	h (days)
Female (share)	−0.0013	0.00459	0.776	1.000	302,823	129
Date of birth (years)	−0.143	0.0592	0.016	0.151	238,086	104
Formal labor market entry (years)	−0.00517	0.0223	0.817	1.000	244,029	103
Date of separation (years)	−0.011	0.0223	0.622	1.000	246,357	104
Age (years)	0.125	0.0513	0.015	0.151	283,453	125
Days since account opened	−2.12	1.01	0.035	0.276	272,139	116
No CURP (share)	−0.00231	0.00221	0.295	1.000	166,414	71
Previous job duration (weeks)	0.424	0.315	0.178	0.892	199,288	85
Previous job total earnings (2024 MXN)	1,630	844	0.053	0.369	197,020	83
Previous job monthly wage (2024 MXN)	311	214	0.147	0.879	241,684	102

Note: Sharp RD (reduced form) estimates for each predetermined covariate, with a local linear polynomial, triangular kernel, optimal bandwidth (Calonico et al., 2020) and robust bias-corrected p-values (Calonico et al., 2014). The Holm column corrects for the ten simultaneous tests; under that adjustment no covariate retains significance at the 5 percent level.

5 Results

This section presents the empirical results of the fuzzy regression discontinuity design (RDD) used to estimate the causal effect of the *Retiro Parcial por Desempleo* (RPD) program on labor market outcomes. I begin by documenting the first-stage effects of program eligibility on take-up, which justifies the fuzzy RDD strategy. I then estimate the local average treatment effect of RPD take-up on job search behavior and medium-term labor market outcomes. The results are complemented by a heterogeneity analysis that explores how these effects vary by gender, income level, age, and exposure to the COVID-19 pandemic.

5.1 First Stage

This section presents evidence that workers eligible to the RPD program (with at least two years of social security contributions at the time of job displacement) are significantly more likely to take up than ineligible workers. These first-stage results confirm three key points: (i) the running variable has been correctly constructed, (ii) the first stage of the fuzzy regression discontinuity design is strong, and (iii) eligibility has a meaningful, albeit modest, effect on program take-up.

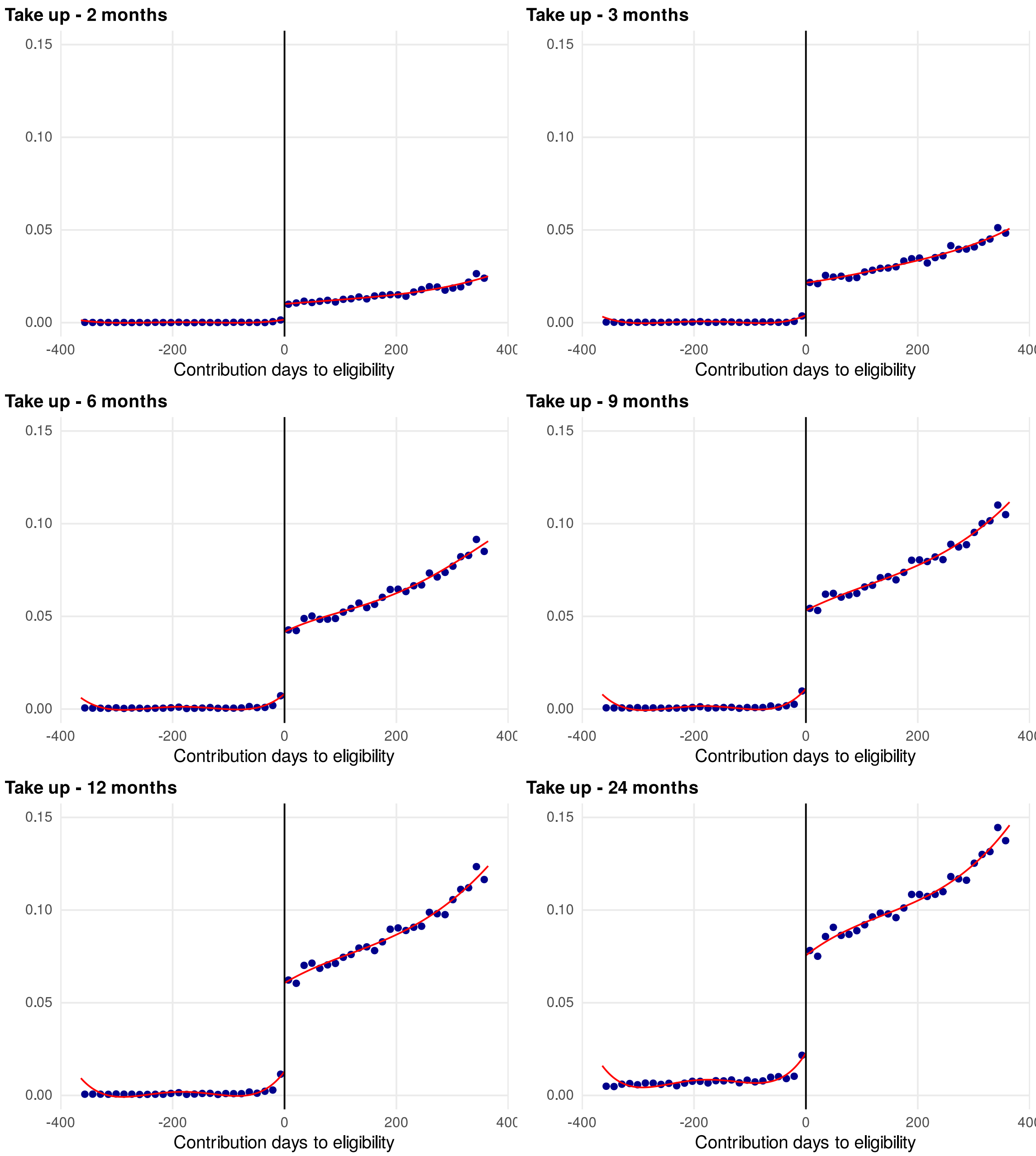
Figure 7 shows that displaced workers just above the eligibility threshold are more likely to access the RPD program to withdraw funds from their pension account. Crossing the threshold raises the probability of using the program within twelve months by 3.6 percentage points (standard error 0.3), estimated on 81,709 observations inside an optimal bandwidth of ±35 days. The first stage is strong: the t statistic of 11.5 corresponds to an F statistic of 133, far above conventional weak-instrument thresholds, so the second-stage estimates are not contaminated by a weak instrument. Inference from Figure 8 and Table 3 confirms that the take-up discontinuity rises from 0.8 percentage points two months after displacement to 3.6 points at twelve months, and stabilizes near 4.0 points by month 24.

Interestingly, Figure 7 shows that take-up increases among initially ineligible workers after the twelfth month following displacement. This pattern should not be interpreted as conventional non-compliance. Rather, it reflects a change in eligibility status over time. Under Eligibility Mode B, workers become eligible for the RPD program once they accumulate at least five years of participation in the formal labor market, regardless of their contribution history (see Section 2 and Figure 1). Given that the analytic sample is restricted to individuals displaced between their third and fourth year in the formal sector, these workers only reach the five-year threshold approximately one year after displacement. Once eligible, they take up the program at rates comparable to the original compliers, which accounts for the observed stabilization in the intent-to-treat (ITT) effect after month 12 (see Figure 8).

Table 3 confirms that the ITT effect on program take-up is highly statistically significant, albeit small in magnitude. This justifies the use of a fuzzy RDD strategy to estimate the local average treatment effect of RPD on labor market outcomes among compliers. The modest size of the first-stage effect is consistent with the institutional design of the RPD program: unlike traditional unemployment insurance schemes, RPD allows only a partial withdrawal from workers' own pension accounts. It is, by design, a self-financed benefit that internalizes the cost of unemployment support.

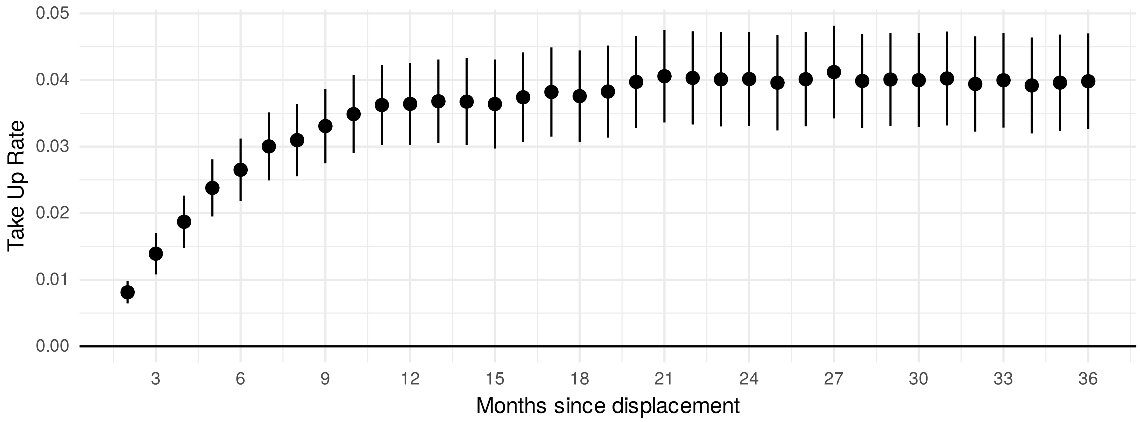
It is worth pausing on what that magnitude implies for everything that follows, because the correct reading of every estimate below depends on it. In a fuzzy design the first stage is also the share of compliers: 3.6 percentage points means that of every hundred workers near the threshold, roughly four use the program *because* they crossed the eligibility rule and would not have used it otherwise. The LATE reported below describes that group — not the average worker in the sample, and not the average user of the program. Two consequences follow. Effects will look large relative to population means precisely because they are concentrated in a small and specific subset; and reduced-form effects, which average over the entire population near the threshold, will be correspondingly small (Table A.2).

Figure 7: Eligibility Effect on Take Up



Note: This figure displays (i) sample means calculated within evenly spaced 14-day bins relative to the eligibility threshold, and (ii) the estimated regression function obtained using a local linear polynomial with a triangular kernel and optimal bandwidth selection following the method of Calonico et al. (2015). The results show a discrete increase in program take-up at the eligibility threshold.

Figure 8: Dynamic Effect of Eligibility on Take Up



Note: This figure presents the estimated coefficients of interest from (1). Estimates are obtained using a local linear regression with a triangular kernel and optimal bandwidth selection. Bias-corrected point estimates and robust standard errors are computed following the method of Calonico et al. (2014). Vertical lines across point estimates represent 95% confidence intervals.

Table 3: Eligibility Effect: First Stage

	Take up - 2 months	Take up - 3 months	Take up - 6 months	Take up - 9 months	Take up - 12 months
RPD	0.008	0.014	0.027	0.033	0.036
	(0.001)	(0.002)	(0.002)	(0.003)	(0.003)
Mean at cutoff - ineligible	0.00174	0.00602	0.0133	0.0187	0.0231
Observations	877,749	877,749	877,749	877,749	877,749

Note: This table reports the estimated coefficient of interest from Equation 4.1, obtained using a local linear regression with a triangular kernel and optimal bandwidth selection. Bias-corrected point estimates and robust standard errors are computed following the procedure of Calonico, Cattaneo, and Titiunik (2014).

5.2 Job Search Outcomes

I now turn to the effects of RPD take-up on labor market outcomes, estimated using the fuzzy regression discontinuity design described in (2). Figure 9 and Table 4a present the dynamic treatment effects on unemployment duration and employment survival. By three months after displacement, taking up RPD increases the probability of remaining unemployed (i.e., survival in unemployment) by 30.7 percentage points ($p = 0.036$). After 36 months, the cumulative effect on unemployment duration amounts to 36.3 additional weeks ($p = 0.022$, 95% CI 5.2–67.4). These effects are statistically significant at the 5% level and align with the broader empirical literature: income support during unemployment typically prolongs joblessness. The additional weeks amount to roughly 70% of the 51.5 weeks that ineligible workers spend out of formal employment over the same horizon.[5] These findings underscore the potential social costs of income-substitu-

[5] That percentage is a reference for scale, not a rate, and the two quantities it compares are not additive. The 51.5-week base is the mean for *all* ineligible workers, while the 36.3-week effect belongs to compliers — a group composed, by construction, of workers who withdraw once they become eligible, and whose spells are plausibly

tion programs such as RPD, particularly through the lens of forgone contributions to the social security system due to extended non-employment spells.

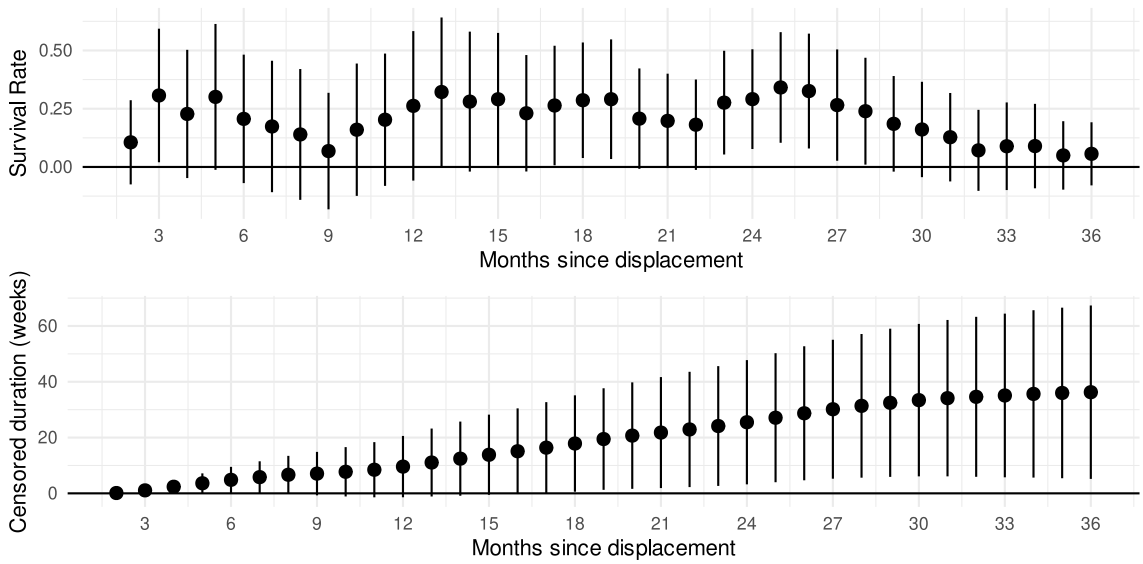
For completeness, Appendix Figure A.1, Figure A.2, and Figure A.3, along with Table A.2, present intention-to-treat (ITT) estimates for the same outcomes, as described in (1), as well as the corresponding RD plots. The ITT estimates show smaller effects than those reported in comparable international settings. For instance, Britto (2022) finds that a lump-sum unemployment benefit program in Brazil increased employment survival by 1.9 percentage points after three months, whereas the estimated effect of RPD is just 0.7 percentage points over the same period. Similarly, Velázquez Guadarrama (2023), using ENOE survey data, reports that eligibility to RPD increases unemployment duration by 11.8 weeks (or 82.85 days), while the ITT estimate from this study points to a much smaller increase of just 0.65 weeks over a three-year horizon. Importantly, none of the ITT estimates reported here are statistically significant at the 5% level.

Turning to employment quality, Table 4b reports the estimated effects of RPD take-up on post-displacement outcomes such as earnings and job duration. The composition of re-employed workers appears similar around the eligibility threshold in terms of prior earnings (see column 4), suggesting no substantial selection effects. However, there is limited evidence that RPD take-up improves subsequent job quality. The point estimate for monthly earnings in the next job is sizeable — \$9,429 MXN, against a mean of \$8,327 among ineligible workers — but it does not reach significance at the 5% level ($p = 0.053$). The effects on next job cumulative earnings and job duration are statistically indistinguishable from zero. The robust finding is the absence of a systematic improvement rather than the presence of a wage gain.

When placed in context with the existing literature, these ITT results point to a mixed picture. Britto (2022) finds a statistically significant 1.7% decline in annual earnings associated with Brazil's program, while eligibility for RPD is associated with a non-significant 4.6% increase. Velázquez Guadarrama (2023), using ENOE data, finds a smaller yet statistically significant increase of 2.47% in wage income among eligible workers. Overall, while some point estimates suggest potentially positive effects of RPD on earnings, the lack of statistical significance and the inconsistency with prior studies call for cautious interpretation.

longer to begin with. The same caution applies to the survival estimate: 30.7 percentage points cannot be added to the 77.8% of ineligible workers still unemployed at three months to obtain a probability.

Figure 9: Dynamic Effect of RPD on Survival and Duration of Unemployment



Note: This figure presents the estimated coefficients of interest from (2). Estimates are obtained using a local linear regression with a triangular kernel and optimal bandwidth selection. Bias-corrected point estimates and robust standard errors are computed following the method of Calonico et al. (2014). Vertical lines across point estimates represent 95% confidence intervals.

Table 4: RPD Effect: Job Search Outcomes

(a) Survival and Duration of Unemployment

	Survival - 3 months	Survival - 36 months	Duration - 6 months (wks)	Duration - 36 months (wks)
RPD	0.31	0.06	4.85	36.27
	(0.15)	(0.07)	(2.37)	(15.86)
Mean at cutoff - ineligible	0.778	0.106	20.5	51.5
Observations	877,749	877,749	877,749	877,749

(b) Next Job Quality

	Monthly earnings	Total earnings	Job duration (weeks)	Prev. job earnings
RPD	9428.61	67095.54	16.49	9626.59
	(4882.68)	(61137.52)	(14.54)	(6055.86)
Mean at cutoff - ineligible	8,327	87,306	35.7	9,005
Observations	873,434	873,434	873,434	873,434

Note: This table reports the estimated coefficient of interest from Equation 4.2, where treatment is defined as program take-up 12 months after displacement. Estimates are obtained using a local linear regression with a triangular kernel and optimal bandwidth selection. Bias-corrected point estimates and robust standard errors are computed following the method of Cattaneo, Idrobo, and Titiunik (2024). Currencies are expressed in 2024 Mexican Pesos.

5.3 Medium Term Effects

This section examines the medium-term labor market outcomes associated with participation in the RPD program. Table 5 reports the estimated effects for RPD users, computed using the fuzzy regression discontinuity design described in (2). None of the estimates reach statistical significance at the 5% level. The point estimates are negative for both cumulative earnings ($ −20,379 MXN over three years, p = 0.64) and months worked (-1.4 months, p = 0.59), which is consistent with adjustment on the extensive margin, but the standard errors are large enough that these should be read as an absence of evidence for improvement rather than as evidence of decline.

That absence is itself the substantive result. Taken together with the job-search and next-job estimates, these findings indicate that the RPD program extends the duration of unemployment while failing to enhance post-displacement job quality or long-term earnings in the formal sector — contrary to the predictions of the more optimistic unemployment insurance literature (Acemoglu and Shimer 1999; 2000), and consistent with evidence from comparable Latin American schemes, where lump-sum severance in Brazil also prolongs formal non-employment without systematically improving subsequent earnings (Britto 2022), and where the wage gains associated with unemployment insurance in Argentina are modest relative to the duration cost (González-Rozada and Ruffo 2023).

Table 5: RPD Effect: Earnings, Employment and Wages over Medium Term

(a) Earnings

	Total 3 years	Year 1	Year 2	Year 3
RPD	−20378.63	−13116.15	−16418.20	−3121.18
	(44061.00)	(12983.56)	(17775.58)	(24304.58)
Mean at cutoff - ineligible	148,762	31,965	54,471	62,345
Observations	877,749	877,749	877,749	877,749

(b) Months Worked

	Total 3 years	Year 1	Year 2	Year 3
RPD	−1.41	−1.95	−0.57	−0.15
	(2.65)	(1.26)	(1.12)	(1.42)
Mean at cutoff - ineligible	15.3	3.8	5.67	5.86
Observations	877,749	877,749	877,749	877,749

(c) Average Monthly Earnings

	Total 3 years	Year 1	Year 2	Year 3
RPD	903.31	634.12	−1806.90	−259.16
	(1808.21)	(1814.44)	(1954.97)	(2146.99)
Mean at cutoff - ineligible	8,903	8,012	8,959	9,902
Observations	794,688	578,415	650,874	633,586

Note: This table reports the estimated coefficient of interest from Equation 4.2, where treatment is defined as program take-up 12 months after displacement. Estimates are obtained using a local linear regression with a triangular kernel and optimal bandwidth selection. Bias-corrected point estimates and robust standard errors are computed following the method of Cattaneo, Idrobo, and Titiunik (2024). Currencies are expressed in 2024 Mexican Pesos.

5.4 Additional Results: Heterogeneous Treatment Effects

In this section, I present additional results on the heterogeneity of the effect of the RPD program on labor market outcomes. I explore the heterogeneity of the treatment effect by age, wage, and exposure to COVID-19 in Table 7, Table 8, and Table 10, respectively.

Each split is estimated on disjoint subsamples, so each subgroup has its own optimal bandwidth and its own estimate. Comparing the magnitude of two such estimates does not establish that the subgroups differ: a large coefficient in one half and a small one in the other is consistent with a common underlying effect estimated with different precision. I therefore report, alongside each split, a formal test of the difference between the two halves on the 36-month duration outcome. Because the subsamples are disjoint, the variance of the difference is the sum of the variances. The results are summarized in Table 6: the gradients by prior income and by sex are statistically distinguishable, while those by age and by COVID-19 exposure are not, and I read them accordingly throughout.

Table 6: Test of the difference in the 36-month duration effect between the two halves of each split

Split	Difference (wks)	Std. err.	p	Verdict
Previous income (below vs. above median)	50.8	(23.3)	0.029	Distinguishable
Age (below vs. above median)	32.1	(24.5)	0.190	Not distinguishable
Sex (men vs. women)	57.8	(25.2)	0.022	Distinguishable
COVID-19 exposure (pre-pandemic vs. full)	30.8	(33.7)	0.361	Not distinguishable

Note: Difference between the two subgroup estimates of the effect of RPD take-up on unemployment duration 36 months after displacement. Subsamples are disjoint, so the standard error of the difference is the square root of the sum of the subgroup variances.

Income. Table 7 explores heterogeneous effects of the RPD program by pre-displacement income, dividing the sample at the median wage in the previous job ($6,669 MXN). Take-up rates do not differ significantly between the above- and below-median wage groups, suggesting comparable levels of program engagement across income levels.

The impact on labor market outcomes, however, does differ. Among workers earning below the median, RPD take-up increases unemployment duration over the 36-month horizon by 46 weeks (p = 0.007), while the estimate for the above-median group is small and negative and cannot be distinguished from zero. Crucially, the difference between the two groups is itself statistically significant (p = 0.029), so this is a genuine gradient rather than a contrast of imprecise magnitudes.

In terms of reemployment outcomes, RPD recipients with above-median pre-displacement wages secure jobs paying $6,417 MXN more per month (p = 0.035); this is the only reemployment-wage gain in any subgroup that reaches significance at the 5% level. No statistically significant effect is found on the wages of reemployed workers in the below-median group. Over the full three-year horizon neither group shows a detectable change in cumulative earnings: the below-median point estimate is a loss of roughly $39,836 MXN and the above-median estimate a gain, but both are far from conventional significance (p = 0.30 and 0.81).

These findings highlight the importance of accounting for income heterogeneity in program design. The duration cost of the RPD falls on workers who arrive at unemployment with less saving and thinner support networks, and the only detectable improvement in reemployment pay accrues to those who were already better paid. The structure of the benefit may therefore reinforce existing inequalities rather than offset them.

Table 7: RPD Effect: Heterogeneity over Wage

(a) Below Median Wage

	Take up (First stage)	Survival - 3 months	Duration - 36 months (wks)	Job duration (wks)	Monthly Earnings	Total Earnings - 3 years	Months Worked - 3 years	Monthly Earnings - 3 years
RPD	0.04	0.38	46.16	15.80	4737.20	−39836.31	−3.64	−60.70
	(0.00)	(0.14)	(17.22)	(21.26)	(5196.56)	(38497.50)	(3.41)	(1467.20)
Mean at cutoff - ineligible	0.0121	0.797	53.8	36.2	7,269	119,239	14.6	7,617
Observations	438,875	438,875	438,875	436,925	436,925	438,875	438,875	393,137

(b) Above Median Wage

	Take up (First stage)	Survival - 3 months	Duration - 36 months (wks)	Job duration (wks)	Monthly Earnings	Total Earnings - 3 years	Months Worked - 3 years	Monthly Earnings - 3 years
RPD	0.04	−0.05	−4.67	17.05	6416.72	17516.18	0.66	2081.56
	(0.00)	(0.14)	(15.71)	(21.47)	(3044.42)	(73028.49)	(4.21)	(2808.53)
Mean at cutoff - ineligible	0.027	0.763	49.7	35.2	9,437	178,004	16	10,180
Observations	438,874	438,874	438,874	436,509	436,509	438,874	438,874	401,551

Note: This table reports the estimated coefficient of interest from Equation 4.1, obtained using a local linear regression with a triangular kernel and optimal bandwidth selection. Bias-corrected point estimates and robust standard errors are computed following the procedure of Calonico, Cattaneo, and Titiunik (2014). Panel A reports results for individuals whose earnings in their previous job were below the sample median wage ($6,669), while Panel B reports results for those above the median. All monetary values are expressed in 2024 Mexican Pesos.

Age. Table 8 examines heterogeneous effects of the RPD program by age at the time of displacement, dividing the sample at the median age of 22.9 years. As with income, take-up rates do not differ significantly between younger and older workers, indicating comparable engagement with the program across age groups.

The point estimates point in the same direction as the income gradient: over 36 months, the cumulative increase in unemployment duration is 35 weeks for younger workers against 3 weeks for older ones. That contrast should not, however, be read as a finding. The younger-worker estimate reaches significance only at the 10% level ($p = 0.051$), and the difference between the two age groups is not distinguishable from zero ($p = 0.190$). The data are equally consistent with a genuinely larger effect among younger workers and with a common effect estimated less precisely in the older subsample.

There is likewise no clear evidence of differential effects on post-displacement job quality. All estimates related to wages and job duration in the next job are statistically indistinguishable from zero for both age groups, as are the effects on cumulative three-year earnings ($p = 0.63$ and 0.88).

Age therefore remains a plausible but unproven dimension of heterogeneity here. The pattern is suggestive of greater early-career vulnerability, and is worth pursuing with a design better powered to detect it, but this sample cannot establish that the cost of the RPD falls disproportionately on younger workers.

Table 8: RPD Effect: Heterogeneity over Age

(a) Below Median Age

	Take up (First stage)	Survival - 3 months	Duration - 36 months (wks)	Job duration (wks)	Monthly Earnings	Total Earnings - 3 years	Months Worked - 3 years	Monthly Earnings - 3 years
RPD	0.04	0.23	35.16	−4.43	60.02	−20965.39	−2.11	357.61
	(0.00)	(0.17)	(17.98)	(18.77)	(1805.32)	(42939.90)	(3.82)	(1261.62)
Mean at cutoff - ineligible	0.0162	0.767	47.7	33.4	7,747	141,633	15.9	8,327
Observations	423,857	423,857	423,857	422,155	422,155	423,857	423,857	388,767

(b) Above Median Age

	Take up (First stage)	Survival - 3 months	Duration - 36 months (wks)	Job duration (wks)	Monthly Earnings	Total Earnings - 3 years	Months Worked - 3 years	Monthly Earnings - 3 years
RPD	0.04	0.10	3.10	28.18	9759.34	−9224.03	−1.35	114.08
	(0.00)	(0.12)	(16.58)	(19.68)	(6752.33)	(59252.06)	(3.75)	(2246.94)
Mean at cutoff - ineligible	0.0214	0.793	56.1	38.4	9,032	157,792	14.9	9,601
Observations	423,857	423,857	423,857	421,318	421,318	423,857	423,857	378,703

Note: This table reports the estimated coefficient of interest from Equation 4.2, where treatment is defined as program take-up 12 months after displacement. Estimates are obtained using a local linear regression with a triangular kernel and optimal bandwidth selection. Bias-corrected point estimates and robust standard errors are computed following the method of Cattaneo, Idrobo, and Titiunik (2024). Panel A shows the results for the below median age group (22.9 years), while panel B shows the results for the above median age group. Currencies are expressed in 2024 Mexican Pesos.

Gender. Table 9 examines the heterogeneous effects of the RPD program by gender. Panel A presents results for males, while Panel B reports estimates for females. Take-up rates are similar across groups, with no statistically significant gender differences in program participation 12 months after displacement.

The effects on labor market outcomes do diverge markedly by gender, and here the divergence is testable. Among men, RPD take-up raises cumulative unemployment duration by 49 weeks over the subsequent three years (p = 0.007). Among women the point estimate is negative (-8.7 weeks) and imprecisely estimated. The difference between the two is itself statistically significant (p = 0.022), making sex, alongside prior income, one of the two dimensions along which the cost of the program demonstrably concentrates.

Earnings outcomes tell a more restrained story than the duration results might suggest. Male recipients show a positive point estimate on monthly earnings in their next job ($11,252 MXN), but it reaches only the 10% level (p = 0.099), and their cumulative three-year earnings effect is economically negligible and statistically absent ($3,074 MXN, p = 0.95, against a standard error of \$52,415). Female recipients show no significant gain in monthly earnings and no detectable change in cumulative earnings either (p = 0.79). Neither group, in other words, is compensated for the additional time spent out of formal employment.

The gender gradient admits an interpretation worth making explicit rather than omitting. The effect concentrates among men, who in this sample have stronger attachment to formal employment. Among women the estimator is negative and insignificant, which is consistent with a portion of their post-displacement transitions running not into job search but out of the formal labor force altogether — a margin these data cannot distinguish. Uniform benefit structures may therefore yield unequal outcomes, but the mechanism behind the gap is not settled by these estimates.

Table 9: RPD Effect: Heterogeneity over Gender

(a) Males

	Take up (First stage)	Survival - 3 months	Duration - 36 months (wks)	Job duration (wks)	Monthly Earnings	Total Earnings - 3 years	Months Worked - 3 years	Monthly Earnings - 3 years
RPD	0.04	0.40	49.08	14.55	11252.23	3074.41	−2.73	1066.12
	(0.00)	(0.14)	(18.27)	(16.88)	(6816.09)	(52415.38)	(3.08)	(1827.85)
Mean at cutoff - ineligible	0.0136	0.753	47	33.3	8,288	155,075	16	8,965
Observations	546,893	546,893	546,893	544,534	544,534	546,893	546,893	502,724

(b) Females

	Take up (First stage)	Survival - 3 months	Duration - 36 months (wks)	Job duration (wks)	Monthly Earnings	Total Earnings - 3 years	Months Worked - 3 years	Monthly Earnings - 3 years
RPD	0.04	−0.17	−8.71	15.66	1994.00	−16126.79	1.51	1194.77
	(0.01)	(0.14)	(17.36)	(22.98)	(2160.69)	(59431.83)	(3.68)	(2252.93)
Mean at cutoff - ineligible	0.0281	0.823	59.3	39.7	8,403	137,174	14.1	8,794
Observations	330,856	330,856	330,856	328,900	328,900	330,856	330,856	291,964

Note: This table reports the estimated coefficient of interest from Equation 4.2, where treatment is defined as program take-up 12 months after displacement. Estimates are obtained using a local linear regression with a triangular kernel and optimal bandwidth selection. Bias-corrected point estimates and robust standard errors are computed following the method of Cattaneo, Idrobo, and Titiunik (2024). Panel A shows the results for the male group, while panel B shows the results for the female group. Currencies are expressed in 2024 Mexican Pesos.

Exposure to COVID-19. Table 10 explores heterogeneous effects of the RPD program based on whether individuals were displaced before or after the onset of the COVID-19 pandemic. Panel A includes workers who were displaced before March 2019, and thus had at least a full year of unemployment before the pandemic hit in March 2020. These individuals are considered to have no exposure to pandemic-related labor market disruptions during their initial post-displacement period. Panel B, by contrast, focuses on individuals displaced after March 2020, who faced full exposure to the labor market conditions created by the pandemic.

Among workers with no exposure to the pandemic (Panel A), RPD take-up produces a substantial and significant increase in unemployment duration: 53 additional weeks over three years (p = 0.015), essentially one additional year of joblessness. Among workers fully exposed to the crisis (Panel B), the estimate is 22 weeks and cannot be distinguished from zero (p = 0.39).

It is tempting to read that contrast as evidence that the program behaves differently in a downturn, but the data do not support the inference: the difference between the two cohorts is itself statistically indistinguishable from zero (p = 0.36). The estimates are equally consistent with a genuinely smaller effect during the crisis and with the same effect measured less precisely in a smaller and more turbulent sample. Neither cohort shows a detectable change in cumulative three-year earnings, though the point estimate for the pandemic cohort is large and positive.

The relevance of this contrast for policy design is nevertheless greater than its statistical strength, and it is worth stating why. The central cost documented in this paper — additional non-employment — is not a fixed property of the program. It measures how much reemployment the liquidity displaces, and how much reemployment is available to displace depends on the state of the labor market. When vacancies are plentiful, the RPD competes with them and its duration cost is real; when the economy is destroying formal jobs at scale, the alternative the worker forgoes is not a faster exit from unemployment but an equally long spell without liquidity, so the same withdrawn peso buys more insurance at less distortion. This is the standard logic of optimal unemployment insurance, in which desirable generosity rises as the behavioral response weakens and the marginal value of liquidity grows (Chetty 2008). The implication is uncomfortable for the current design: if the cost of the program moves with the business cycle and its parameters do not, the RPD is miscalibrated in both states of the world — too permissive when the labor market functions, and too restrictive precisely when its insurance value is highest. Establishing this empirically, rather than by appeal to theory, would require a design with more power in the crisis cohort than this one has.

Table 10: RPD Effect: Heterogeneity over Exposure to COVID-19

(a) No Exposure (Displaced before March 2019)

	Take up (First stage)	Survival - 3 months	Duration - 36 months (wks)	Job duration (wks)	Monthly Earnings	Total Earnings - 3 years	Months Worked - 3 years	Monthly Earnings - 3 years
RPD	0.04	0.37	52.76	18.01	12416.29	−8539.55	−2.92	1241.50
	(0.00)	(0.19)	(21.75)	(26.88)	(8273.95)	(55380.27)	(4.25)	(2301.33)
Mean at cutoff - ineligible	0.0113	0.775	50.7	38.3	8,003	143,580	15.5	8,497
Observations	561,937	561,937	561,937	560,491	560,491	561,937	561,937	503,428

(b) Full Exposure (Displaced after March 2020)

	Take up (First stage)	Survival - 3 months	Duration - 36 months (wks)	Job duration (wks)	Monthly Earnings	Total Earnings - 3 years	Months Worked - 3 years	Monthly Earnings - 3 years
RPD	0.03	0.01	21.99	−9.39	5010.11	37277.66	−2.31	4689.68
	(0.01)	(0.20)	(25.68)	(19.50)	(2769.82)	(69127.25)	(4.50)	(3634.36)
Mean at cutoff - ineligible	0.0491	0.78	48.4	30.1	9,004	169,405	16.1	9,879
Observations	156,826	156,826	156,826	155,207	155,207	156,826	156,826	148,663

Note: This table reports the estimated coefficient of interest from Equation 4.2, where treatment is defined as program take-up 12 months after displacement. Estimates are obtained using a local linear regression with a triangular kernel and optimal bandwidth selection. Bias-corrected point estimates and robust standard errors are computed following the method of Cattaneo, Idrobo, and Titiunik (2024). Currencies are expressed in 2024 Mexican Pesos.

5.5 Longer Unemployment, or Exit into Informality?

The IMSS administrative data observe formal employment and only formal employment. What this paper measures, precisely, is the time until the next registered formal job; a worker who withdraws her savings and then works on her own account or in the informal sector appears in these data exactly as one who remains unemployed and searching. The behavioral interpretation of the main result therefore depends on which of the two stories dominates, and the question is better confronted than deferred.

There are reasons to think the second is not marginal. Britto (2022) shows for Brazil that 57% of the formal employment displaced by an extension of unemployment insurance is offset by informal employment, and Gerard and Gonzaga (2021) document that the efficiency cost of these programs is lower where formal reentry is unlikely to begin with. For Mexico, the evidence points the same way: Carreño Godínez (2025) finds that eligibility for the RPD substantially raises the probability of transitioning into self-employment, consistent with the withdrawal operating as seed capital in the absence of credit. With these data it is not possible to apportion the 36.3 additional weeks between the two margins, and it would be wrong to claim otherwise.

What can be said is that the welfare reading and the retirement reading come apart here, and only the first is ambiguous. If a substantial share of the additional weeks is informal work rather than idle search, the consumption cost of the program is smaller than the duration estimate suggests. The cost to retirement savings, however, is identical either way: neither self-employment nor informal work generates contributions to the retirement subaccount, and neither accrues contribution weeks. If those transitions prove durable — as the self-employment finding suggests they might — the loss would not stop at the weeks estimated here but would extend to a more persistent exit from the contributory pension system altogether. Section 5.6 quantifies that channel.

5.6 What the Additional Unemployment Costs at Retirement

The RPD is not only an income-substitution program; it is mechanically a draw against retirement savings. Because the estimates above identify an effect on time spent outside formal employment, they can be translated into a cost expressed in pension terms, which is the unit in which a worker ultimately bears it. The translation runs through two channels. First, the amount withdrawn stops compounding inside the individual account: if W is the withdrawal and r the real return, the cost at retirement n years later is $W(1+r)^n$. Second, and less visibly, the worker makes no contributions during the additional weeks outside formal employment: if $\bar{w}$ is the prior monthly wage, c the contribution rate to the retirement subaccount, and Δ the additional months of non-employment, that cost is $\bar{w} \cdot c \cdot \Delta \cdot (1+r)^n$.

Calibrating this with the sample's own parameters — an average withdrawal of $7,691 MXN, an average prior wage of $9,095 MXN per month, the 36.3 additional weeks estimated above, the 6.5% contribution rate in force over the estimation window, and 40 years to retirement for a worker separating at the sample mean age of 25 — the total loss in accumulated balance at age 65 ranges from 5.4 to 11.6 times the amount withdrawn, across real returns of 3% to 5%. Of that loss, 39% comes not from the withdrawal itself but from the contributions the additional unemployment prevented; because both channels compound at the same rate, that share does not depend on the return assumption. Propagating the confidence interval of the duration estimate through the second channel, the total at a 4% real return lies between $40,306 and $80,920 MXN.

Two caveats bound the exercise. It is illustrative rather than actuarial: it holds the real return constant, ignores fees, and does not model partial recovery of contributions after reemployment. And it combines a parameter estimated on compliers with means computed over the whole sample, so it describes a worker of average characteristics induced by the eligibility rule rather than any observed individual. Neither caveat disturbs the qualitative conclusion, which is that a liquidity instrument that appears inexpensive at the moment of use — under one month of prior salary, on average — is an order of magnitude more costly measured at retirement, and that a substantial minority of that cost is invisible in the withdrawal itself because it operates through the labor market response this paper estimates.

Conclusions

This thesis evaluates the labor market effects of Mexico's *Retiro Parcial por Desempleo* (RPD), a policy that allows displaced formal sector workers to make early withdrawals from their pension savings. Leveraging a discontinuity at two years of social security contributions, I implement a fuzzy regression discontinuity design to estimate the local average treatment effect of program take-up on labor market outcomes.

The results show that RPD take-up substantially increases the time spent outside formal employment, with significant effects observable as early as three months after displacement and persisting over a three-year horizon. The increase is large and stable across bandwidths, suggesting that liquidity provision through the RPD displaces formal reemployment. However, there is no consistent evidence that RPD take-up improves job quality or medium-term formal labor income. The only reemployment-wage gain significant at the 5% level accrues to workers whose prior earnings were already above the median, and in no subgroup is the effect on three-year cumulative earnings distinguishable from zero. Testing differences between subgroups directly, the duration cost concentrates on lower-paid workers and on men; the analogous gradients by age and by pandemic exposure are suggestive but not statistically distinguishable.

Take-up rates remain low, and the first-stage estimates are small but precise, indicating that the program reaches a narrow set of workers and that the estimates describe that narrow set. This has a consequence beyond the labor market. Because the additional weeks are weeks without contributions to the retirement subaccount, the cost of the program compounds until retirement: as Section 5.6 shows, the loss in accumulated balance is several times the sum withdrawn, and a substantial share of it comes not from the withdrawal itself but from the unemployment the program induces. That conclusion is robust to the interpretive ambiguity discussed in Section 5.5, since neither informal employment nor self-employment generates contributions or accrues contribution weeks.

Several limitations of the study should be acknowledged. First, the fuzzy RD design identifies local effects for compliers—those induced to take up the program by marginal eligibility—limiting external validity. Second, the administrative data do not capture informal employment or non-wage welfare outcomes, potentially understating total adjustment margins. Third, despite large point estimates, wide confidence intervals in the second stage warrant cautious interpretation. Fourth, because workers who are ineligible at displacement become eligible under Mode B once they reach five years in the formal sector, part of the control group is treated within the three-year window; the long-horizon estimates reported here should therefore be read as a lower bound.

My results underscore the institutional tensions in Mexico's fragmented income protection system. Severance pay, the de jure main income substitute, is often inaccessible due to weak legal enforcement (Sadka et al. 2024; Degetau 2023). In this context, the RPD emerges as a second-best option: it is self-financed, administratively simple, and unlikely to induce fraud. Yet this model shifts the burden of income protection onto workers themselves—especially those with fewer savings—thereby reinforcing labor market inequalities.

A more ambitious policy agenda would involve developing a publicly funded unemployment insurance system that ensures risk-sharing beyond individual savings. However, such a system would require overcoming institutional barriers: effective delivery depends not only on eligibility design but also on credible enforcement and administrative capacity.

The findings point to several promising directions for future research. First, aggregate behavioral effects—such as changes in employer hiring or firing practices—remain an open question. Second, the observed heterogeneity by income, age, gender, and macroeconomic context calls for richer models of household and labor market behavior under liquidity constraints. Finally, the intersection between legal institutions and social protection policy in low-capacity states remains an underexplored but vital area for understanding the design and implementation of unemployment assistance in developing countries.

Appendix

A Descriptive statistics

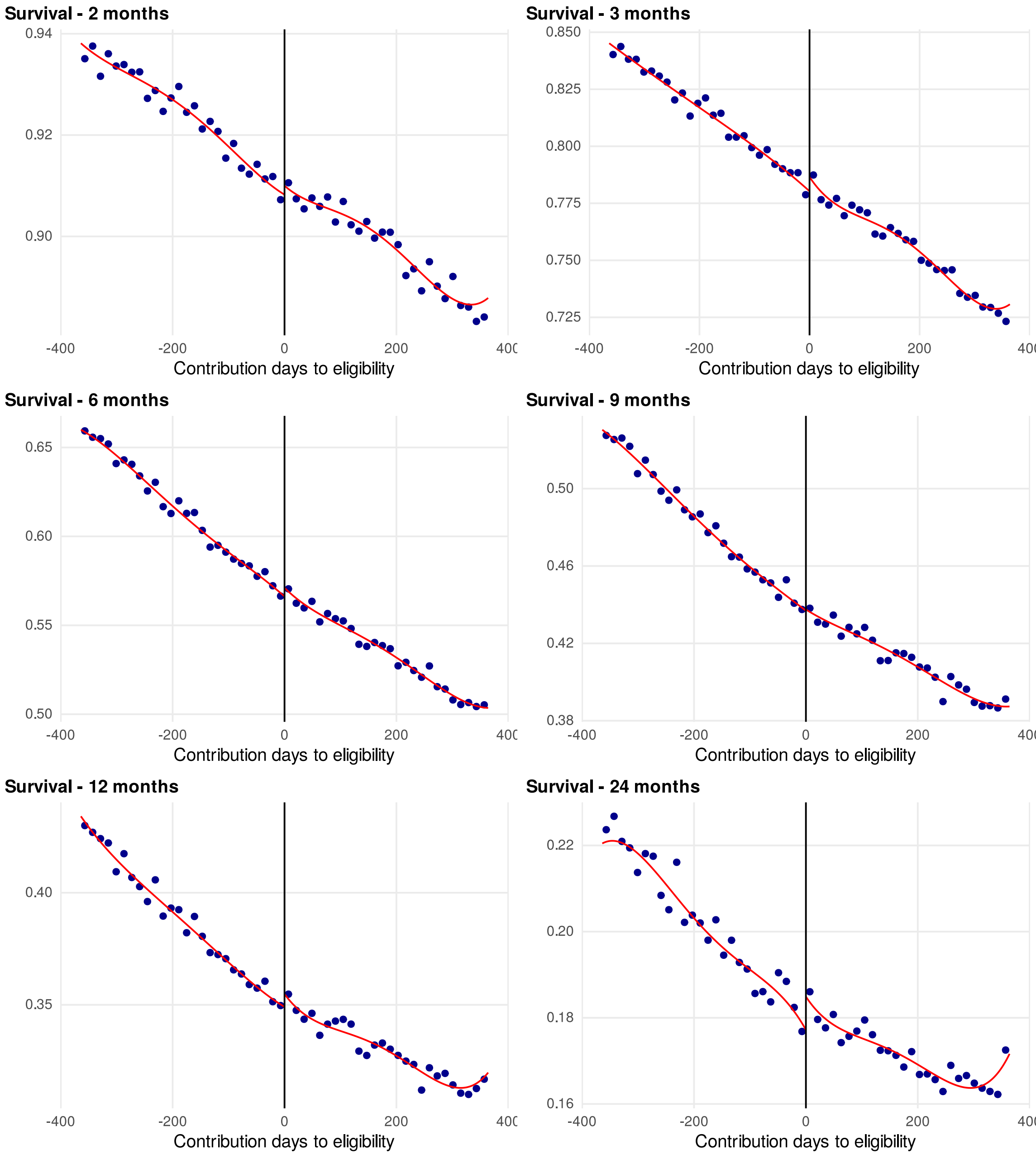
B Reduced-form estimates

Table A.1: Summary Statistics

Variable	n	Mean	Median	Std.Dev.	Min	Max
Covariates						
Age at displacement date	847,714	24.7	22.9	5.5	18	65
Birth date	847,714	1,993.4	1,994.6	5.7	1,948.5	2,003.5
Date worker began working	877,749	2,014.6	2,014.9	2.1	2,010	2,018
Days since account opened at displacement date	877,749	1,264.5	1,259	103.1	1,095	1,460
Displacement date	877,749	2,018	2,018.3	2.1	2,013	2,022
Female	877,749	0.4	0	0.5	0	1
No CURP	877,749	0	0	0.2	0	1
Previous Job						
Duration of previous job (weeks)	877,749	27.7	16.4	29.7	0.1	156.3
Monthly earnings in previous job (2024 MXN)	877,749	9,095.4	6,669.2	19,424.5	1,631.2	3,634,843.3
Total earnings in previous job (2024 MXN)	877,749	54,235.8	25,769.4	82,490.9	107.6	2,894,843.6
Partial Withdrawal (RPD)						
Amount withdrawn (2024 MXN)	62,632	7,690.6	6,637.5	4,365.9	324.2	134,480.5
Balance at withdrawal (2024 MXN)	62,632	21,435.4	19,983.8	10,838.8	470.9	1,171,354.5
Contributed weeks at take up date	62,632	125.9	130	29.1	0	1,273
Days to take up	62,632	440.8	228	477.2	46	2,892
Days withdrawn	62,632	318.4	301	126.8	0	4,075
Job Search Outcomes						
Survival out of formal employment after 3 months	877,749	0.8	1	0.4	0	1
Survival out of formal employment after 36 months	877,749	0.1	0	0.3	0	1
Weeks out of formal employment censored at 3 months	877,749	12.1	12.9	1.8	6.3	12.9
Weeks out of formal employment censored at 36 months	877,749	53.1	32.6	49.1	6.3	154.3
Next Job Quality						
Duration of next job (weeks)	873,434	36.8	14.7	59.3	0.1	611.3
Monthly earnings in next job (2024 MXN)	873,434	8,461.7	6,948.8	6,643.8	1,599.3	2,171,267.3
Total earnings in next job (2024 MXN)	873,434	91,888.4	23,412.3	245,995	105.4	10,500,054.5
Medium Term Outcomes (Over 3 Years)						
Average monthly earnings after displacement (2024 MXN)	794,688	9,007.6	7,515.1	5,841.1	3,163.5	84,996
Months worked after displacement	877,749	15.2	14.1	11.5	0	89.5
Total earnings after displacement (2024 MXN)	877,749	149,523.5	102,486.9	180,264.6	0	4,035,597.8

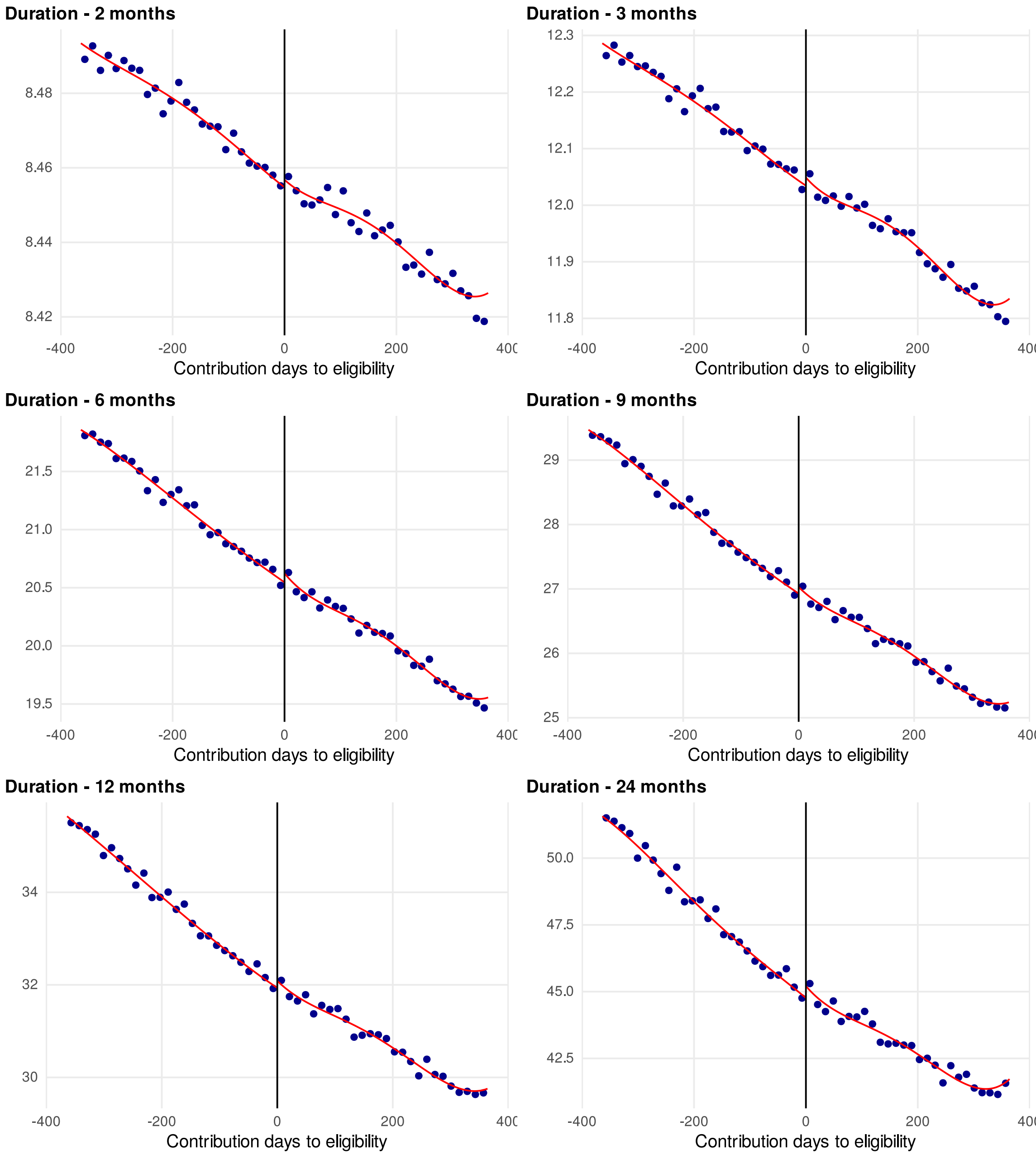
Note: This table reports summary statistics for the variables used in the analysis. The statistics are based on the final sample used for estimation.

Figure A.1: Eligibility Effect: Survival in Unemployment



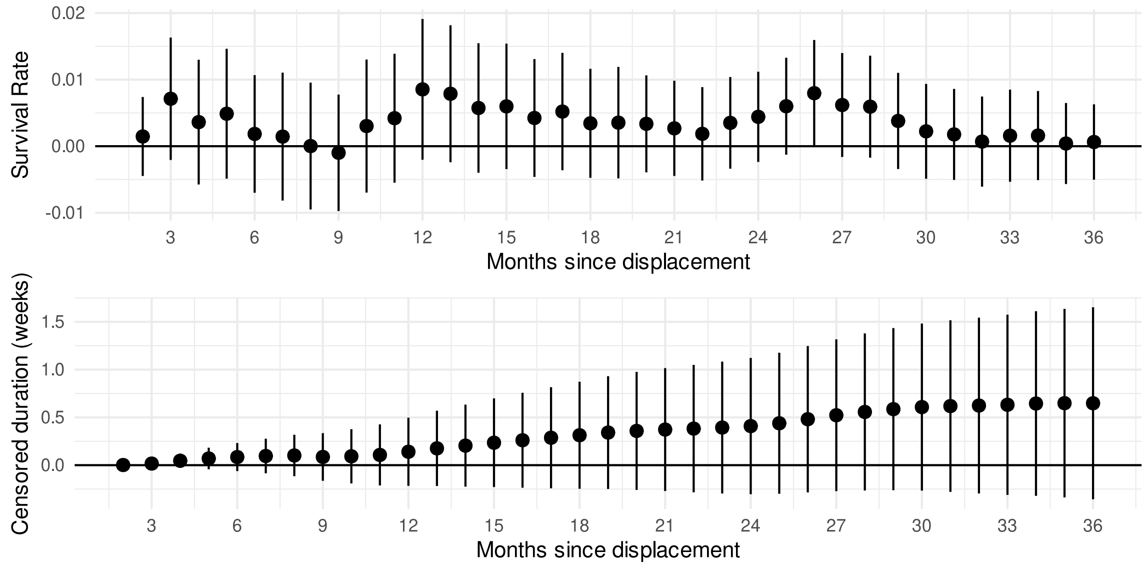
Note: This figure shows (*i*) sample means for evenly-spaced 14-day bins relative to the eligibility threshold, and (*ii*) the underlying regression function estimated using a triangular kernel, optimal bandwidth and a linear local polynomial using Calonico et al. (2015).

Figure A.2: Eligibility Effect: Duration of Unemployment



Note: This figure shows (*i*) sample means for evenly-spaced 14-day bins relative to the eligibility threshold, and (*ii*) the underlying regression function estimated using a triangular kernel, optimal bandwidth and a linear local polynomial using Calonico et al. (2015).

Figure A.3: Dynamic Effect of Eligibility on Survival and Duration of Unemployment



Note: This figure shows the estimated coefficient of interest in (1). The estimates are obtained using a triangular kernel, optimal bandwidth and a linear local polynomial. Bias corrected estimates and robust standard errors computed using Calonico et al. (2014). 95 percent confidence intervals.

Table A.2: Eligibility Effect: Job Search Outcomes

(a) Survival and Duration of Unemployment

	Survival - 3 months	Survival - 36 months	Duration (wks) - 6 months	Duration (wks) - 36 months
RPD	0.007	0.001	0.085	0.648
	(0.005)	(0.003)	(0.075)	(0.513)
Mean at cutoff - ineligible	0.78	0.108	20.5	51.9
Observations	877,749	877,749	877,749	877,749

(b) Next Job Quality

	Monthly earnings (2024 MXN)	Total earnings (2024 MXN)	Job duration (weeks)	Prev. job earnings
RPD	283.48	1469.40	0.56	276.07
	(163.04)	(2178.74)	(0.62)	(215.71)
Mean at cutoff - ineligible	8,333	87,703	35.7	8,978
Observations	873,434	873,434	873,434	873,434

Note: This table reports the estimated coefficient of interest from Equation 4.1, obtained using a local linear regression with a triangular kernel and optimal bandwidth selection. Bias-corrected point estimates and robust standard errors are computed following the procedure of Calonico, Cattaneo, and Titiunik (2014).

B.1 Reduced-Form Estimates and Their Relation to the LATE

Table A.2 reports the effect of *eligibility* rather than of use — the reduced form, or intent-to-treat. No reduced-form effect on a labor market outcome is significant at the 5% level, and this is exactly what a first stage of 3.6 percentage points implies. An effect of 36.3 weeks concentrated on compliers dilutes to well under a week once averaged over the entire population near the threshold, the overwhelming majority of whom never use the program. The reduced form is therefore not in tension with the LATE; it is the same evidence expressed per capita rather than per user.

One caveat is worth stating for readers who attempt the division directly. The ratio of the reduced form to the first stage does not exactly reproduce the LATE reported in Table 4, because each fuzzy estimate selects its own optimal bandwidth and applies its own bias correction, so numerator and denominator are not computed on the same effective sample. The stability of the LATE with respect to the bandwidth is examined in Table A.3 below.

C Sensitivity to the Bandwidth

Table A.3 re-estimates the two principal effects with the bandwidth fixed at multiples of the optimum, scaling the bias-correction bandwidth in the same proportion. Point estimates are positive across the entire range and of comparable magnitude between half and one times the optimal width; at wide bandwidths they attenuate, as expected once the local polynomial approximation absorbs observations increasingly far from the threshold. Statistical significance at the 5% level concentrates around the optimal bandwidth, reflecting the usual bias–variance trade-off in this class of design.

Table A.3: Sensitivity of the Main Effects (LATE) to the Bandwidth

Bandwidth	h (days)	LATE	Std. err.	p	Eff. N
Survival in unemployment at 3 months (pp)					
$0.5 \times h$	21	46.4	33.5	0.166	50,761
$0.75 \times h$	32	38.0	18.9	0.045	74,570
Optimal h	43	30.7	14.6	0.036	100,511
$1.5 \times h$	64	18.3	10.7	0.086	150,065
$2 \times h$	85	12.4	8.6	0.150	201,604
Unemployment duration at 36 months (weeks)					
$0.5 \times h$	24	41.7	33.0	0.206	55,428
$0.75 \times h$	36	40.4	20.0	0.044	84,099
Optimal h	48	36.3	15.9	0.022	112,213
$1.5 \times h$	71	20.8	11.6	0.073	168,821
$2 \times h$	95	12.1	9.4	0.195	225,261

Note: Fuzzy RD estimates re-computed with the bandwidth fixed at multiples of the optimum, scaling the bias-correction bandwidth proportionally. Robust bias-corrected p-values. The optimal-bandwidth row may differ marginally from the main tables because the bandwidth is set explicitly rather than selected internally.

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